Jack Reiman

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Education

University of Washington **Sep. 2025 – Present**
Mechanical Engineering | GPA: 3.51 *Seattle, WA*

Summary of Qualifications

Software: ANSYS, SOLIDWORKS, Microsoft 365, Autodesk Inventor, Autodesk Fusion 360, Onshape, KeyShot
Manufacturing: CNC Routing & Milling, Additive Manufacturing (FDM & Formlabs SLA), OMAX Waterjet, UW Composite Shop Trained

Relevant Experience

R&D Team Member **Sep. 2025 – Present**
Advanced Robotics @ UW *Seattle, WA*

- Engineered a lightweight omni wheel using Toray 3900 prepreg carbon fiber plates; validated load cases in ANSYS with a FOS > 2
- Designed custom polyurethane rollers and silicone molds using SLA master models to eliminate wheel scrubbing and diameter mismatch
- Designed a 19:1 low backlash cycloidal gearbox for DJI M3508 motors to replace bulky stock planetary gearboxes for turret yaw control
- Prototyped eccentric shafts and cycloidal discs using Formlabs Tough 2000 resin on a Form 3 SLA printer
- Developed and maintained a standardized SOLIDWORKS Hole Wizard & Advanced Hole library used team wide
- Designed and manufactured a custom robot controller integrating keyboard and joystick inputs within a 3D printed housing

Instructor **May 2026 – Present**
Quantum Robotics *Redmond, WA*

- Instructed middle school competition teams in Onshape CAD modeling, including 2D drawings and 3D assemblies
- Taught VEXcode C++ programming concepts, including sensor integration and simple control algorithms
- Guided students through the engineering design process and technical notebook documentation

Team Lead **Aug. 2021 – May 2025**
VEX V5 Robotics Team 61002G *Ellensburg, WA*

- Led the design and construction of robots, winning VEX Worlds Division Champion (2025) among 11,000+ teams
- Utilized SOLIDWORKS for mechanical prototyping and planning
- Wrote a custom Stanley controller using C++ for consistent autonomous control

Personal Projects

Differential Swerve Module **May 2025**
Personal Project *Ellensburg, WA*

- Designed and manufactured a custom differential swerve module
- Applied my knowledge of gear design to design compact bevel gears and optimize packaging

Leverless Gamecube Controller **November 2025**
Personal Project *Seattle, WA*

- Designed a custom Melee controller from scratch in SOLIDWORKS, FDM printing the housing and keycaps
- Hand wired Kailh hotswap sockets with Gateron Yellow switches to a Raspberry Pi Pico running open source firmware